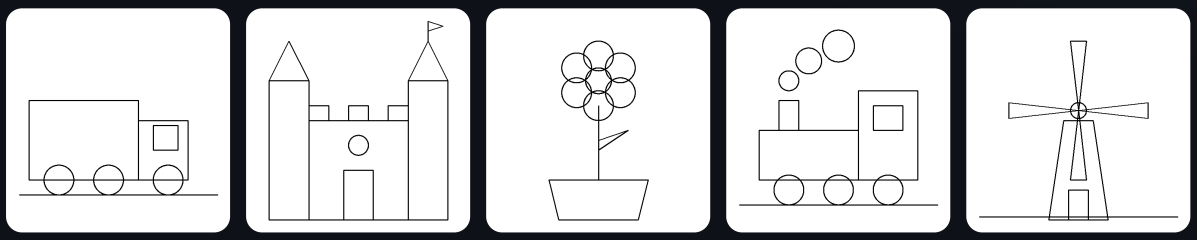
CMPN322 · MID TERM WRITTEN EXAM · PRACTICE SET 2

Computer Graphics
New Practice Exams

Five new mid-term models (11–15), different from the ten earlier models: new numbers, new figures, new theory items. Same layout as the Fall 2025 paper, full model answers, Question 3 in your notebook method (0–20 grid, drawRect, drawCircle).

Part 1 · 5 exam papers | Part 2 · Program template | Part 3 · Model answers | Part 4 · Answer key



5	25	40	5
Exam models	MCQ	Theory items	Programs

Contents

Solve the paper first, then open its model answer

01	**Part 1 — Exam Papers** · 5 models, questions only	04
02	**Part 2 — Program Template** · the fixed part of Question 3	14
03	**Part 3 — Model Answers**	17
04	**Part 4 — Quick Answer Key**	44

MODEL	Q3 FIGURE	QUESTION 2	PAPER	ANSWER
M11	Truck	Midpoint ellipse · P′ = T(5, 2) · R(−90) · T(−5, −2) · P	04	17
M12	Castle	Midpoint circle · P′ = R(45) · T(−1, 4) · S(3, 3) · P	06	23
M13	Flower	Bresenham line · P′ = T(4, 4) · S(−1, 1) · S(2, 2) · T(−4, −4) · P	08	29
M14	Train	Midpoint ellipse · P′ = T(2, 3) · R(60) · T(−2, −3) · T(6, 0) · P	10	34
M15	Windmill	DDA line · P′ = S(1, −1) · T(0, 5) · R(180) · P	12	39

What is new here, and what comes from the references

- **Mine (practice):** the numbers in the problems, the five figures of Question 3, and the choice of questions.
- **From the references:** every method, formula, definition, comparison and correction — each answer shows its file and page.
- Question 3 uses your notebook method: gluOrtho2D(0, 20, 0, 20) with your own drawRect and drawCircle.

حل كل نموذج في ساعة من غير ما تبص على الحل، وبعدين صحّح من Part 3.

PART 1 · EXAM PAPERS

Exam Papers

Models 11–15 in the layout of the real mid-term: Question 1 (MCQ + theory), Question 2 (algorithm + composite transformation), Question 3 (program). One hour each.

حل كل نموذج في ساعة من غير ما تفتح الحل.

CG / EXAM PAPER / MODEL 11 · model answer on page 17

Subject: (CMPn322) Computer graphics **Academic Year:** 2025-2026 **Semester:** Fall

Branch: Computer **Time:** 1 hour **Academic Level:** Senior 1

Questions FOR THE MID TERM WRITTEN EXAM — Practice Model 11

Number of Pages: 2 **Number of Questions:** 3 **Attempt all questions**

Question1 , [LO's 1,2,3,4] (5 Points uniformly distributed)

a) Choose the correct answer:

1) A true color image has a color depth of ------

A) 8 bit B) 16 bit

C) 24 bit D) 32 bit

2) If we use 7 bits pixel value in a 24-bit lookup table, the lookup table has:

A) 128 entries, 384 bytes B) 128 entries, 24 bytes

C) 7 entries, 168 bytes D) 24 entries, 128 bytes

3) Window (W.l, W.r, W.b, W.t) = (0, 40, 0, 20), viewport (V.l, V.r, V.b, V.t) = (100, 300, 50, 150). For (XW, YW) = (10, 5), (XV, YV) = ?

A) (50, 25) B) (150, 100)

C) (150, 75) D) (75, 150)

4) A Liquid Crystal Display (LCD) is an example of ------ displays.

A) Emissive B) Non-emissive

5) DDA line drawing produces the following first points for a line with endpoints (10, 10) and (14, 13):

A) (11,11) (12,11) (13,12) B) (11,11) (12,12) (13,12) C) (11,10) (12,11) (13,12)

b) Answer the following:

1. Define **Normalized coordinates** · **Graphics pipeline**.
2. Compare Differentiate between **Geometric model and mathematical model**.
3. True / False Put (✓) or (✗), and correct the false statements:
    1. Rasterization is cutting vertices that will not be on the screen. ( )
    2. A frame buffer that supports true color has 24 planes of memory (8 for each color). ( )
    3. A scan line is one dot of the raster. ( )
4. Complete
    1. Pick input primitive is _____.
    2. In gray-scale images 8 bits/pixel produce _____ gray levels.

Practice Model 11 — Question 2 & Question 3

Question2 [LO's 2, 3] (10 Points)

1) Determine pixel positions along the ellipse path in the first quadrant using midpoint ellipse algorithm. Given: input ellipse parameter $r_x = 4$ and $r_y = 6$

- $p1_0 = r_y^2 - r_x^2 r_y + \frac{1}{4} r_x^2$
- $p1_{k+1} = p1_k + 2r_y^2 x_{k+1} + r_y^2 \ (p1_k \le 0) \quad | \quad p1_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_y^2 \ (p1_k > 0)$
- $p2_0 = r_y^2 (x_0 + \frac{1}{2})^2 + r_x^2 (y_0 - 1)^2 - r_x^2 r_y^2$
- $p2_{k+1} = p2_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_x^2 \ (p2_k \le 0) \quad | \quad p2_k - 2r_x^2 y_{k+1} + r_x^2 \ (p2_k > 0)$

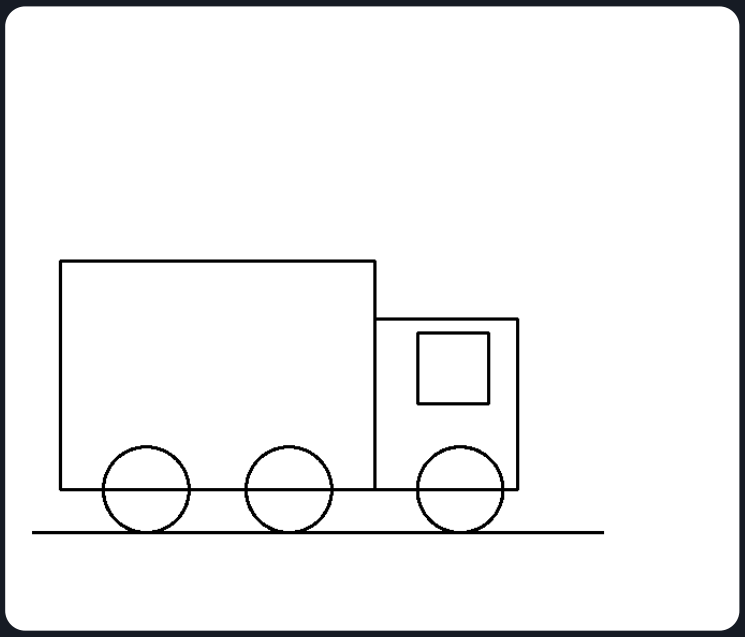
2) A programmer applies the following composite transformation to every vertex of a 2D object:

$$P' = T(5, 2) \cdot R(-90) \cdot T(-5, -2) \cdot P$$

Deconstruct the Transformation: Break down the sequence of operations. Describe what happens to the object step-by-step. (5 Points)

Question 3 [LO's 3,4] (5 Points)

Write a program to draw the following fig



CG / EXAM PAPER / MODEL 12 · model answer on page 23

Subject: (CMPn322) Computer graphics **Academic Year:** 2025-2026 **Semester:** Fall

Branch: Computer **Time:** 1 hour **Academic Level:** Senior 1

Questions FOR THE MID TERM WRITTEN EXAM — Practice Model 12

Number of Pages: 2 **Number of Questions:** 3 **Attempt all questions**

Question1 , [LO's 1,2,3,4] (5 Points uniformly distributed)

a) Choose the correct answer:

1) ------ is cutting vertices that will not be on the screen.

A) Rasterization B) Clipping

C) Vertex shader D) Fragment shader

2) A 640 × 480 display uses a frame buffer with 8 bits per pixel (no LUT). Its size is:

A) 2,457,600 bytes B) 921,600 bytes

C) 38,400 bytes D) 307,200 bytes

3) The matching viewport for a world window with R = 2 when the screen window is 400 × 400 is:

A) glViewport(0, 0, 200, 400) B) glViewport(0, 0, 400, 200)

C) glViewport(0, 0, 400, 400) D) glViewport(0, 0, 800, 400)

4) glutInitWindowPosition gives an initial location for the ------ corner of the display window.

A) lower left B) upper right

C) lower right D) upper left

5) Which equation calculates the next pixel of the DDA line from (2, 9) to (6, 1)?

A) (x − 1, y − m) B) (x + 1, y + m)

C) (x − 1/m, y − 1) D) (x + 1/m, y + 1)

b) Answer the following:

1. Define **True color** · **Antialiasing**.
2. Compare Differentiate between **Window and viewport**.
3. True / False Put (✓) or (✗), and correct the false statements:
    1. glViewport sets the world window. ( )
    2. By default the viewport is the entire screen window. ( )
    3. Bresenham's algorithm uses floating-point arithmetic. ( )
4. Complete
    1. In DDA, if |m| ≤ 1: x_{k+1} = _____ and y_{k+1} = _____.
    2. Vector graphics print precisely even when they are _____.

Practice Model 12 — Question 2 & Question 3

Question2 [LO's 2, 3] (10 Points)

1) Given a circle radius r = 13, determine positions along the circle octant in the first quadrant from x = 0 to x = y using the midpoint circle algorithm.

- Initial point (0, r) & $P0 = 1 - r$
- $p_k < 0$: $y_{k+1} = y_k$, $P_{k+1} = P_k + 2x_{k+1} + 1$
- $p_k \geq 0$: $y_{k+1} = y_k - 1$, $P_{k+1} = P_k + 2x_{k+1} + 1 - 2y_{k+1}$

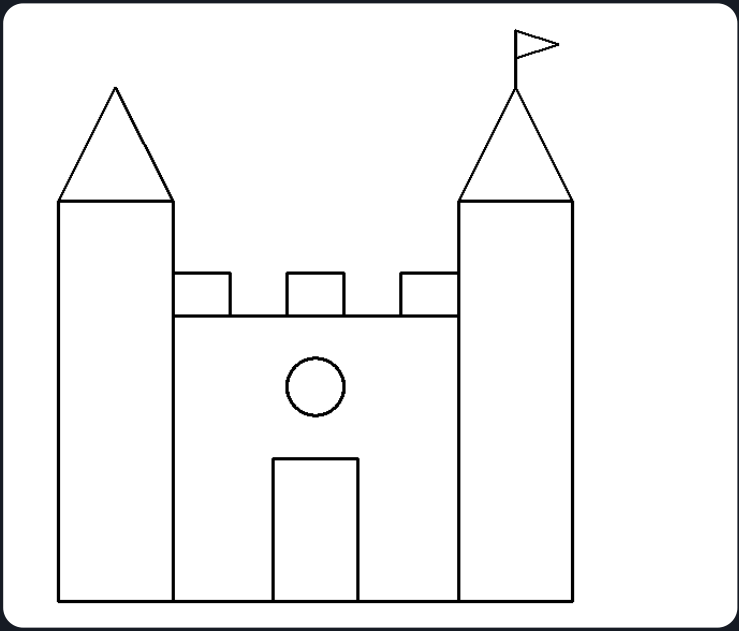
2) A programmer applies the following composite transformation to every vertex of a 2D object:

$$P' = R(45) \cdot T(-1, 4) \cdot S(3, 3) \cdot P$$

Deconstruct the Transformation: Break down the sequence of operations. Describe what happens to the object step-by-step. (5 Points)

Question 3 [LO's 3,4] (5 Points)

Write a program to draw the following fig



CG / EXAM PAPER / MODEL 13 · model answer on page 29

Subject: (CMPn322) Computer graphics **Academic Year:** 2025-2026 **Semester:** Fall

Branch: Computer **Time:** 1 hour **Academic Level:** Senior 1

Questions FOR THE MID TERM WRITTEN EXAM — Practice Model 13

Number of Pages: 2 **Number of Questions:** 3 **Attempt all questions**

Question1 , [LO's 1,2,3,4] (5 Points uniformly distributed)

a) Choose the correct answer:

1) A ------ device has the input primitive: coordinate position (x, y).

A) Locator B) Valuator

C) Choice D) String

2) With direct coding of RGB using 2 bits per primary color, the number of colors is:

A) 6 B) 8

C) 64 D) 512

3) Window (0, 50, 0, 50), viewport (0, 500, 0, 250). A viewport point is (200, 100). Its world coordinates are:

A) (20, 20) B) (20, 10)

C) (40, 20) D) (2000, 500)

4) In region 2 of the midpoint ellipse algorithm we ------

A) increment x, calculate y B) increment y, calculate x C) decrement y, calculate x

5) Midpoint circle with r = 6: the first three points after (0, 6) are:

A) (1,6) (2,5) (3,5) B) (1,6) (2,6) (3,5) C) (1,5) (2,5) (3,4)

b) Answer the following:

1. Define **Input primitives** · **Homogeneous coordinates**.
2. Compare Differentiate between **Region 1 and region 2 of the midpoint ellipse**.
3. True / False Put (✓) or (✗), and correct the false statements:
    1. A point (x, y) on an ellipse in one quadrant gives the ellipse points in the other three quadrants. ( )
    2. The inverse rotation $R^{-1}(\theta) = R(\theta)$. ( )
    3. In a simple color frame buffer each bit plane drives an individual color gun. ( )
4. Complete
    1. To scale about a point that is not the origin we use translate – _____ – translate.
    2. In a raster scan system the _____ takes care of the actual display process.

Practice Model 13 — Question 2 & Question 3

Question2 [LO's 2, 3] (10 Points)

1) Use Bresenham's line drawing algorithm to digitize the line with endpoints (5, 5) and (17, 12). Show the decision parameter at each step and plot the pixels.

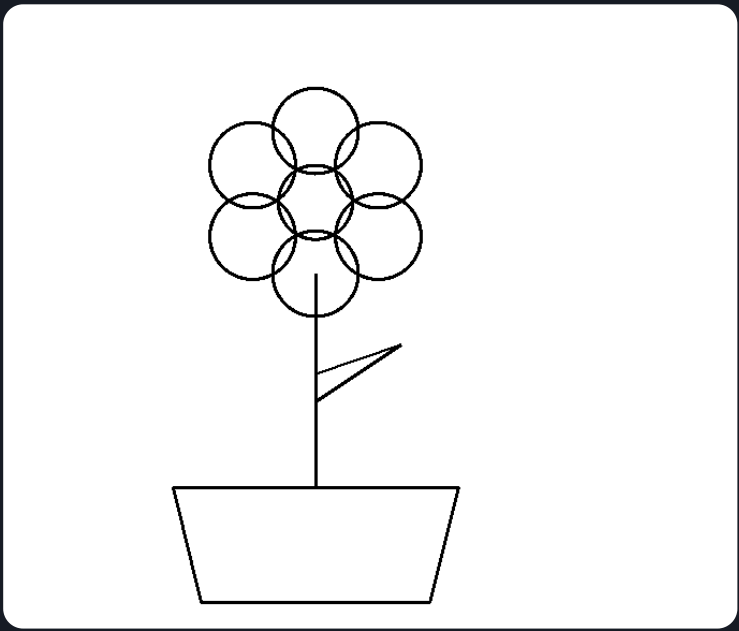
2) A programmer applies the following composite transformation to every vertex of a 2D object:

$$P' = T(4, 4) \cdot S(-1, 1) \cdot S(2, 2) \cdot T(-4, -4) \cdot P$$

Deconstruct the Transformation: Break down the sequence of operations. Describe what happens to the object step-by-step. (5 Points)

Question 3 [LO's 3,4] (5 Points)

Write a program to draw the following fig



CG / EXAM PAPER / MODEL 14 · model answer on page 34

Subject: (CMPn322) Computer graphics **Academic Year:** 2025-2026 **Semester:** Fall

Branch: Computer **Time:** 1 hour **Academic Level:** Senior 1

Questions FOR THE MID TERM WRITTEN EXAM — Practice Model 14

Number of Pages: 2 **Number of Questions:** 3 **Attempt all questions**

Question1 , [LO's 1,2,3,4] (5 Points uniformly distributed)

a) Choose the correct answer:

1) Shear (skewing) deforms the shape like ------

A) a mirror image B) uniform scaling C) shifted slices

2) In gray-scale images, 6 bits/pixel produce ------ gray levels.

A) 6 B) 64

C) 36 D) 256

3) Window (−50, 50, −25, 25), viewport (0, 200, 0, 100). The world point (0, 0) maps to:

A) (0, 0) B) (50, 100)

C) (200, 100) D) (100, 50)

4) The inverse of the scaling $S(s_x, s_y)$ is:

A) $S(-s_x, -s_y)$ B) $S(s_y, s_x)$ C) $S(1/s_x, 1/s_y)$

5) Bresenham line drawing from (0, 0) to (8, 3) gives the first points:

A) (1,1) (2,1) (3,2) B) (1,0) (2,1) (3,1) C) (1,0) (2,0) (3,1)

b) Answer the following:

1. Define **Display device** · **Color depth**.
2. Compare Differentiate between **Vertex shader and fragment shader**.
3. True / False Put (✓) or (✗), and correct the false statements:
    1. Scaling preserves the angles between lines only with uniform scaling $s_x = s_y$. ( )
    2. In old raster scan systems the electron beam is swept across the screen one row at a time. ( )
    3. The midpoint circle algorithm computes one quadrant and gets the rest by symmetry. ( )
4. Complete
    1. Geometric transformation functions: translation, scaling, rotation, _____ and _____.
    2. Diagonal reflection: x′ = _____, y′ = _____.

Practice Model 14 — Question 2 & Question 3

Question2 [LO's 2, 3] (10 Points)

1) Determine pixel positions along the ellipse path in the first quadrant using midpoint ellipse algorithm. Given: input ellipse parameter $r_x = 6$ and $r_y = 2$

- $p1_0 = r_y^2 - r_x^2 r_y + \frac{1}{4} r_x^2$
- $p1_{k+1} = p1_k + 2r_y^2 x_{k+1} + r_y^2 \ (p1_k \le 0) \quad | \quad p1_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_y^2 \ (p1_k > 0)$
- $p2_0 = r_y^2 (x_0 + \frac{1}{2})^2 + r_x^2 (y_0 - 1)^2 - r_x^2 r_y^2$
- $p2_{k+1} = p2_k + 2r_y^2 x_{k+1} - 2r_x^2 y_{k+1} + r_x^2 \ (p2_k \le 0) \quad | \quad p2_k - 2r_x^2 y_{k+1} + r_x^2 \ (p2_k > 0)$

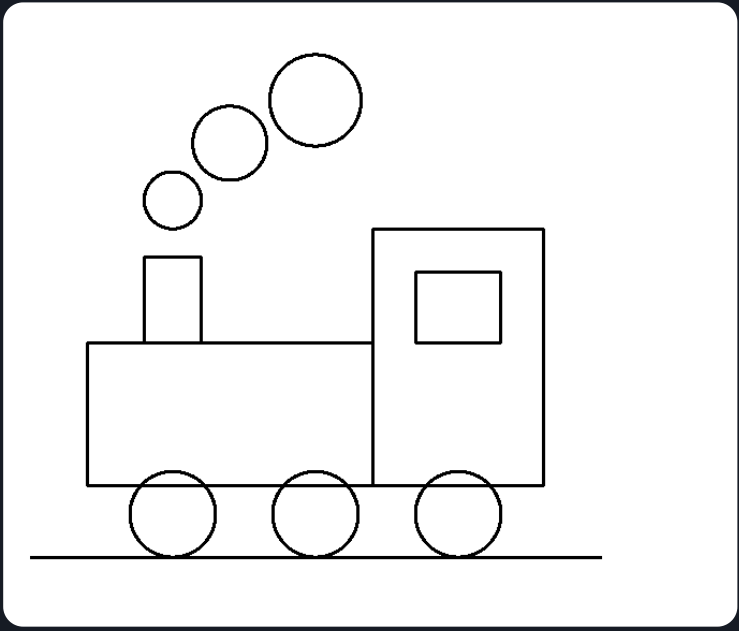
2) A programmer applies the following composite transformation to every vertex of a 2D object:

$$P' = T(2, 3) \cdot R(60) \cdot T(-2, -3) \cdot T(6, 0) \cdot P$$

Deconstruct the Transformation: Break down the sequence of operations. Describe what happens to the object step-by-step. (5 Points)

Question 3 [LO's 3,4] (5 Points)

Write a program to draw the following fig



CG / EXAM PAPER / MODEL 15 · model answer on page 39

Subject: (CMPn322) Computer graphics **Academic Year:** 2025-2026 **Semester:** Fall

Branch: Computer **Time:** 1 hour **Academic Level:** Senior 1

Questions FOR THE MID TERM WRITTEN EXAM — Practice Model 15

Number of Pages: 2 **Number of Questions:** 3 **Attempt all questions**

Question1 , [LO's 1,2,3,4] (5 Points uniformly distributed)

a) Choose the correct answer:

1) Graphics systems convert object descriptions to normalized coordinates to ensure ------

A) faster rendering B) 3D transformations C) device independence

2) If we use 9 bits pixel value in a 16-bit lookup table, the lookup table has:

A) 512 entries, 1024 bytes B) 512 entries, 16 bytes

C) 9 entries, 144 bytes D) 16 entries, 512 bytes

3) Window (10, 30, 10, 30), viewport (0, 100, 0, 100). For (XW, YW) = (20, 25), (XV, YV) = ?

A) (100, 125) B) (75, 50)

C) (50, 75) D) (20, 25)

4) R(30) · R(60) equals:

A) R(90) B) R(1800)

C) R(30) D) R(−30)

5) For an ellipse with $r_x = 6$ and $r_y = 2$, $p1_0$ is:

A) −63 B) 13

C) −59 D) −68

b) Answer the following:

1. Define **Frame rate and refresh rate** · **Composite transformation**.
2. Compare Differentiate between **Direct (Pythagoras) circle method and midpoint circle**.
3. True / False Put (✓) or (✗), and correct the false statements:
    1. A valuator gives a coordinate position (x, y). ( )
    2. $T(t_{x1}, t_{y1}) \cdot T(t_{x2}, t_{y2}) = T(t_{x1} + t_{x2}, t_{y1} + t_{y2})$. ( )
    3. For a line with slope greater than 1 in Bresenham, we interchange the roles of x and y. ( )
4. Complete
    1. glutInitWindowSize(400, 300) creates a display window with a width of _____ and a height of _____.
    2. Supersampling gives more weight to subpixels near the _____ of a pixel area.

Practice Model 15 — Question 2 & Question 3

Question2 [LO's 2, 3] (10 Points)

1) Use the DDA line drawing algorithm to determine the pixel positions along the line from (3, 1) to (7, 10) and plot them.

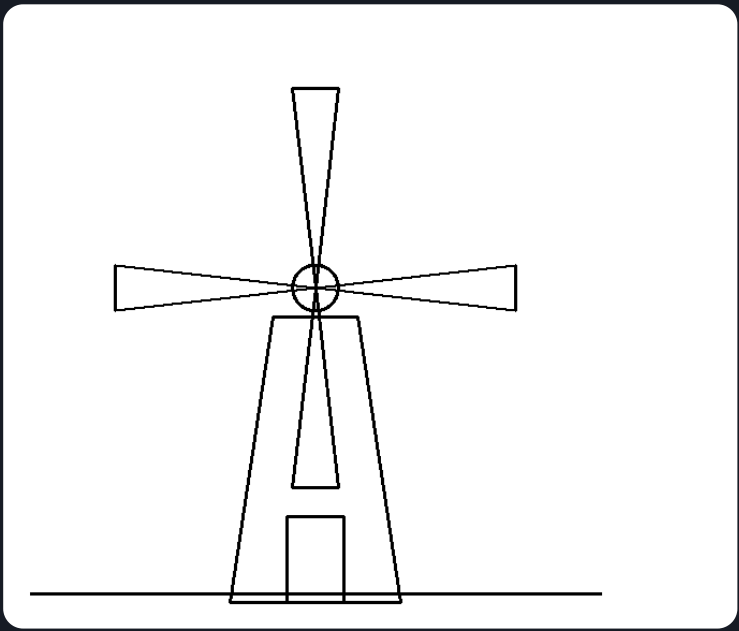
2) A programmer applies the following composite transformation to every vertex of a 2D object:

$$P' = S(1, -1) \cdot T(0, 5) \cdot R(180) \cdot P$$

Deconstruct the Transformation: Break down the sequence of operations. Describe what happens to the object step-by-step. (5 Points)

Question 3 [LO's 3,4] (5 Points)

Write a program to draw the following fig



CG / PART 2 / PROGRAM TEMPLATE

Question 3 — Program Template

Write this once in every answer, then only myDisplay changes

T

T

Source: Lec 3 · p10–11 · Lec 1 · p21 · Lec 5 · p6–8

The fixed part of the program (your notebook method)

ترتيب ورقة الإجابة: الـ drawRect ← includes و main ← myDisplay ← myInit ← drawCircle.

program template START · line 1

```
#include <windows.h>
#include <GL/glut.h>
#include <math.h>
#define PI 3.14159

void drawRect(float x, float y, float w, float h, GLenum mode)
{
    glBegin(mode);
    glVertex2f(x, y);          // lower left
    glVertex2f(x + w, y);      // lower right
    glVertex2f(x + w, y + h);  // upper right
    glVertex2f(x, y + h);      // upper left
    glEnd();
}

void drawCircle(float xc, float yc, float r, int noOfPoints, float shiftAngle, GLenum
mode)
{
    glBegin(mode);
    shiftAngle = shiftAngle * PI / 180;   // degree -> radian
    for (int i = 0; i < noOfPoints; i++)
    {
        float rad = i * 2 * PI / noOfPoints;
        float x = xc + r * sin(rad + shiftAngle);
        float y = yc + r * cos(rad + shiftAngle);
        glVertex2f(x, y);
    }
    glEnd();
}

void myInit()
{
    glClearColor(1.0, 1.0, 1.0, 1.0);   // white background
    glColor3f(0.0, 0.0, 0.0);           // black
    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluOrtho2D(0, 20, 0, 20);           // xmin, xmax, ymin, ymax
}

```

```
// myDisplay() — changes with every figure (see each model answer)

int main(int argc, char** argv)
{
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_RGB);
    glutInitWindowSize(500, 500);
    glutInitWindowPosition(100, 100);
    glutCreateWindow("graphics");
    myInit();
    glutDisplayFunc(myDisplay);
    glutMainLoop();
    return 0;
}
```

END · line 52

PART 3 · MODEL ANSWERS

Model Answers

Full answers for the five models: MCQ with the reason, theory with the reference sentence and page, the algorithm tables, the transformation steps and the program.

صحّح نفسك سؤال سؤال.

CG / MODEL ANSWER / MODEL 11

Model Answer — Model 11

M11

Exam paper on page 04 · every answer shows the reference page of its method

Q1-a

Source: the page is next to each answer

Question 1 (a) — choose the correct answer

#	ANSWER · WHY	SOURCE
1) C	**24 bit** True color image has a color depth of 24 bit (8 bit for each color).	Lec 2 · p24
2) A	**128 entries, 384 bytes** 2^7 = 128 entries · (128 × 24)/8 = 384 bytes	Lec 2 · p39
3) C	**(150, 75)** Sx = 200/40 = 5, Sy = 100/20 = 5 · XV = 100 + 5(10) = 150 · YV = 50 + 5(5) = 75	Lec 3 · p5–6
4) B	**Non-emissive** Non-emissive displays use optical effects to convert light from some other source (LCD); emissive convert electrical energy into light (LED).	Lec 2 · p27
5) B	**(11,11) (12,12) (13,12)** m = 0.75 ≤ 1 · x + 1, y + 0.75: (11, 10.75)→(11, 11) · (12, 11.5)→(12, 12) · (13, 12.25)→(13, 12)	Lec 4 · p19–23

الفكرة المسائل الصغيرة (LUT)، window/viewport، (DDA اتحلّت بنفس قوانين المحاضرة — الأرقام بس جديدة.

Q1-b

Source: the page is next to each answer

Question 1 (b) — theory

1 · Define

Normalized coordinates: To ensure device independence, graphics systems convert object descriptions to normalized coordinates (0 to 1, or −1 to 1). Lec 3 · p2

Graphics pipeline: A conceptual model that describes what steps a graphics system needs to perform to render a 3D scene to a 2D screen. Lec 1 · p24

2 · Compare Geometric model and mathematical model Lec 1 · p9

Geometric model	Mathematical model
Describes how objects appears	Describes how object moves

Attributes: color, texture or shape	A model of a physical process

3 · True / False

1. ✗ **False** Rasterization is cutting vertices that will not be on the screen.
   Correct: That is clipping; rasterization is creating pixels for each part of the object.
   Lec 1 · p24
2. ✓ **True** A frame buffer that supports true color has 24 planes of memory (8 for each color). Lec 2 · p35
3. ✗ **False** A scan line is one dot of the raster.
   Correct: Pixel is one dot of the raster; a scan line is a row of pixels.
   Lec 2 · p25

4 · Complete

1. Pick input primitive is _____. → **selection of a part of the screen** Lec 1 · p14
2. In gray-scale images 8 bits/pixel produce _____ gray levels. → **256** Lec 2 · p23

الفكرة التعريف بجملة المحاضرة، والمقارنة جدول، وأي False لازم تكتب تصحيحه.

Q2-1

Source: Lec 5 · p16–23 · Graphics_Sec 6 · p2

Question 2 (1) — Midpoint ellipse · $r_x = 4$ and $r_y = 6$

HOW TO SOLVE

1. Compute r_x^2, r_y^2 and $p1_0$.
2. Region 1: fill the table (increment x) until $2r_y^2x \geq 2r_x^2y$.
3. Compute $p2_0$ from the last point of region 1.
4. Region 2: fill the table (decrement y) until y = 0.
5. Plot the points.

$r_x^2 = 16$, $r_y^2 = 36$ · initial point (0, 6)

$p1_0 = r_y^2 - r_x^2 r_y + \frac{1}{4}r_x^2 = 36 - 16\times6 + \frac{1}{4}\times16 = -56$

Region 1

Pk_region1	x	y	$2r_y^2x$	$2r_x^2y$
	0	6	0	192
−56	1	6	72	192
52	2	5	144	160
72	3	4	216	128

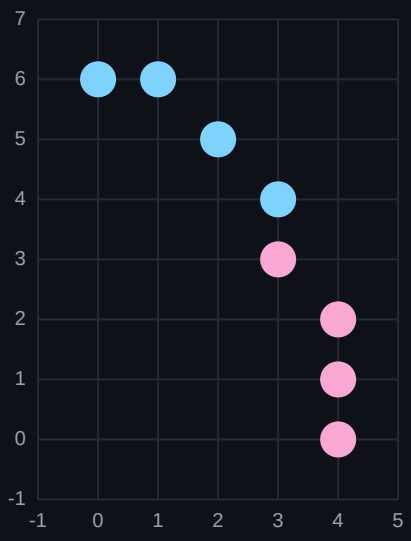
Region 1 stops when $2r_y^2x \geq 2r_x^2y$ → at (3, 4): 216 ≥ 128

Initial point of region 2 = last point of region 1 (3, 4) · $p2_0 = r_y^2(x_0 + \frac{1}{2})^2 + r_x^2(y_0 - 1)^2 - r_x^2r_y^2 = 36\times(3.5)^2 + 16\times(3)^2 - 16\times36 = 9$

Region 2

Pk_region2	x	y	$2r_y{}^2x$	$2r_x{}^2y$
	3	4	216	128
9	3	3	216	96
−71	4	2	288	64
169	4	1	288	32
153	4	0		

Region 1 (blue) · Region 2 (pink)



الفكرة Region 1 بنزوّد x لحد ما $x_y{}^2 2r$ يبقى ≤ $y_x{}^2 2r$، وبعدين Region 2 بنقلّل y لحد الصفر.

Q2-2 Source: Lec 6 · p7, p20–29 · Lec 1 · p22

Question 2 (2) — P′ = T(5, 2) · R(−90) · T(−5, −2) · P

Read from right to left — the matrix nearest to P is applied first:

1. **T(−5, −2):** Translate the object so that the pivot point (5, 2) moves to the origin.
2. **R(−90):** Rotate 90 degree clockwise around the origin (negative angle).
3. **T(5, 2):** Translate the object so that the pivot point is back to its original position.

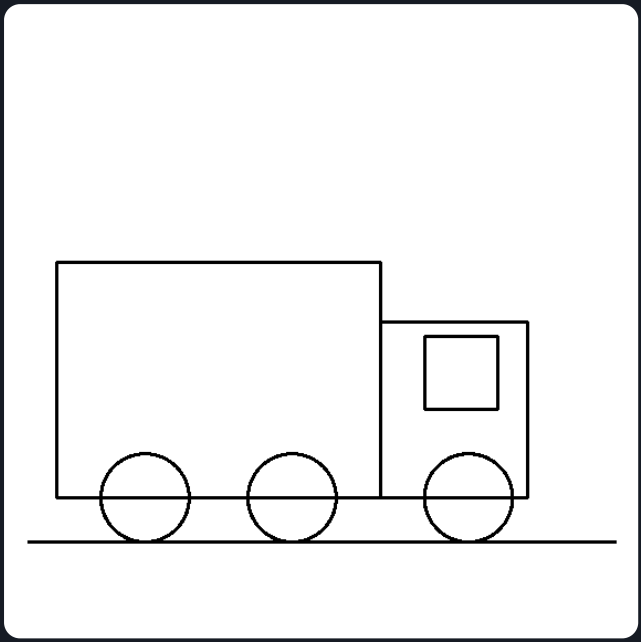
So the object is rotated 90° clockwise around the point (5, 2).

$$\begin{vmatrix} 1 & 0 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} \cos -90 & -\sin -90 & 0 \\ \sin -90 & \cos -90 & 0 \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} 1 & 0 & -5 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{vmatrix}$$

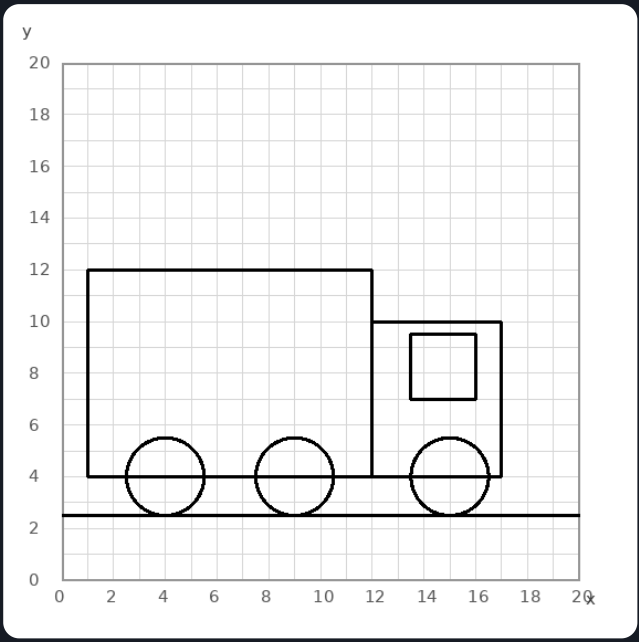
الفكرة اقرا الـ matrices من اليمين للشمال، وكل matrix = جملة بتوصف اللي حصل للشكل.

Q3 Source: main / myInit: Lec 3 · p10 · glBegin / glVertex2f: Lec 3 · p11 · drawCircle (x = xc + r cos θ): Lec 5 · p6–8

Question 3 — program (Truck)



as in the exam



on the 0–20 grid

اكتب الـ includes و drawRect و drawCircle و myInit و main زي صفحة البرنامج الكامل (صفحة 14) — وغيّر بس myDisplay:

myDisplay() START · line 1

```
1  void myDisplay()
2  {
3      glClear(GL_COLOR_BUFFER_BIT);
4      glColor3f(0.0, 0.0, 0.0);
5
6      // cargo box
7      drawRect(1, 4, 11, 8, GL_LINE_LOOP);
8
9      // cab
10     drawRect(12, 4, 5, 6, GL_LINE_LOOP);
11
12     // cab window
13     drawRect(13.5, 7, 2.5, 2.5, GL_LINE_LOOP);
14
15     // wheels
16     drawCircle(4, 4, 1.5, 1000, 0, GL_LINE_LOOP);
17     drawCircle(9, 4, 1.5, 1000, 0, GL_LINE_LOOP);
18     drawCircle(15, 4, 1.5, 1000, 0, GL_LINE_LOOP);
19
20     // road
21     glBegin(GL_LINES);
22     glVertex2f(0, 2.5);
23     glVertex2f(20, 2.5);
24     glEnd();
25
```

```
26      glFlush();
27  }
```

END · line 27

PART	CODE	إزاي جت الأرقام
cargo box	`drawRect(1, 4, 11, 8, …)`	يبدأ من الركن اللي تحت شمال (1, 4) — عرضه 11 وطوله 8 (لحد (12, 12)).
cab	`drawRect(12, 4, 5, 6, …)`	يبدأ من الركن اللي تحت شمال (12, 4) — عرضه 5 وطوله 6 (لحد (17, 10)).
cab window	`drawRect(13.5, 7, 2.5, 2.5, …)`	يبدأ من الركن اللي تحت شمال (13.5, 7) — عرضه 2.5 وطوله 2.5 (لحد (16, 9.5)).
wheels	`drawCircle(4, 4, 1.5, 1000, 0, …)`	دايرة: المركز (4, 4) ونص القطر 1.5.
wheels	`drawCircle(9, 4, 1.5, 1000, 0, …)`	دايرة: المركز (9, 4) ونص القطر 1.5.
wheels	`drawCircle(15, 4, 1.5, 1000, 0, …)`	دايرة: المركز (15, 4) ونص القطر 1.5.
road	`glBegin(GL_LINES) · 2 glVertex2f`	خطوط بالنقط: (0, 2.5) ← (20, 2.5).

الفكرة كل جزء في الصورة = نداء واحد: مستطيل ← drawRect، دايرة ← drawCircle، أي شكل تاني ← بالنقط glBegin.

CG / MODEL ANSWER / MODEL 12

Model Answer — Model 12

M12

Exam paper on page 06 · every answer shows the reference page of its method

Q1-a Source: the page is next to each answer

Question 1 (a) — choose the correct answer

#	ANSWER · WHY	SOURCE
1) B	**Clipping** Clipping: cutting vertices that will not be on the screen.	Lec 1 · p24
2) D	**307,200 bytes** 640 × 480 × 8 / 8 = 307,200 bytes	Graphics_SEC 3 · p2
3) B	**glViewport(0, 0, 400, 200)** W/H = 1 < R → width = W = 400, height = W/R = 200	Lec 3 · p25
4) D	**upper left** glutInitWindowPosition gives an initial location for the upper left corner of the display window.	Lec 1 · p21
5) C	**(x − 1/m, y − 1)** Δx = 4, Δy = −8 → |m| > 1 → step in y; the line goes down → y − 1, x − 1/m	Lec 4 · p22

الفكرة المسائل الصغيرة (LUT)، window/viewport، (DDA اتحلّت بنفس قوانين المحاضرة — الأرقام بس جديدة.

Q1-b Source: the page is next to each answer

Question 1 (b) — theory

1 · Define

True color: A color image with a color depth of 24 bit, 8 bit for each color. Lec 2 · p24

Antialiasing: Methods that compensate for the under sampling process (the jagged, stair-step appearance). Lec 5 · p24

2 · Compare Window and viewport Lec 3 · p3, p8 · Graphics_Sec 4 · p1

Window	Viewport
What is to be displayed	Where it is to be displayed

Region in world coordinates (xwmin, xwmax, ywmin, ywmax)	Region on the display device (xvmin, xvmax, yvmin, yvmax)
gluOrtho2D(left, right, bottom, top)	glViewport(x, y, width, height)

3 · True / False

1. ✗ **False** glViewport sets the world window.
   Correct: gluOrtho2D sets the world window; glViewport sets the viewport.
   Lec 1 · p21 · Lec 3 · p8
2. ✓ **True** By default the viewport is the entire screen window. Lec 3 · p8
3. ✗ **False** Bresenham's algorithm uses floating-point arithmetic.
   Correct: Bresenham uses only incremental integer calculations.
   Lec 4 · p25

4 · Complete

1. In DDA, if $|m| \le 1$: x_{k+1} = _____ and y_{k+1} = _____. → **$x_k + 1 \cdot y_k + m$** Lec 4 · p16
2. Vector graphics print precisely even when they are _____. → **enlarged** Lec 1 · p15

الفكرة التعريف بجملة المحاضرة، والمقارنة جدول، وأي False لازم تكتب تصحيحه.

Q2-1

Source: Lec 5 · p11–13 · Graphics_Sec 6 · p1

Question 2 (1) — Midpoint circle · r = 13

HOW TO SOLVE

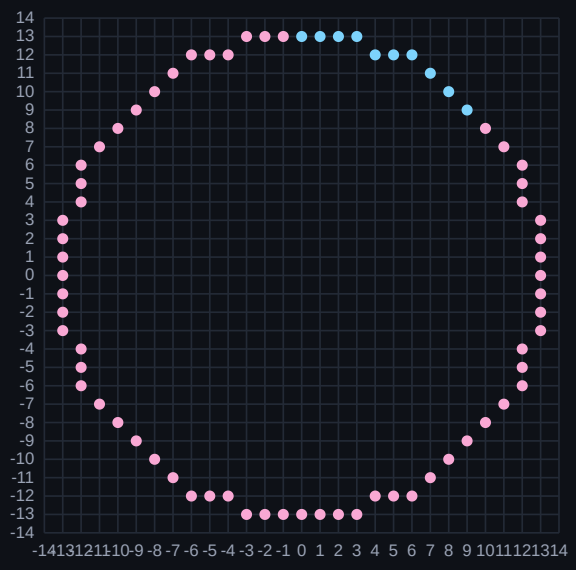
1. P0 = 1 − r, start at (0, r).
2. Use the sign of P to choose y or y − 1, then update P.
3. Stop when x ≥ y.
4. Plot the octant and its symmetry.

Initial point (0, 13) · P0 = 1 − r = -12

P	X	Y	$2x_{k+1}$	$2y_{k+1}$
-	0	13	-	-
−12	1	13	2	26
−9	2	13	4	26
−4	3	13	6	26
3	4	12	8	24
−12	5	12	10	24
−1	6	12	12	24
12	7	11	14	22

Octant points (blue) + eight-way symmetry (pink)

5	8	10	16	20
2	9	9	-	-



الفكرة إشارة P بتحدد y هيفضل ثابت ولا هينزل 1، وبنقف لما x يوصل لـ y.

Q2-2

Source: Lec 6 · p7, p20–29 · Lec 1 · p22

Question 2 (2) — P′ = R(45) · T(−1, 4) · S(3, 3) · P

Read from right to left — the matrix nearest to P is applied first:

1. **S(3, 3):** Scale around the origin: the object becomes 3 times its size.
2. **T(−1, 4):** Translate the object 1 unit to the left and 4 units up.
3. **R(45):** Rotate 45 degree counter-clockwise around the origin.

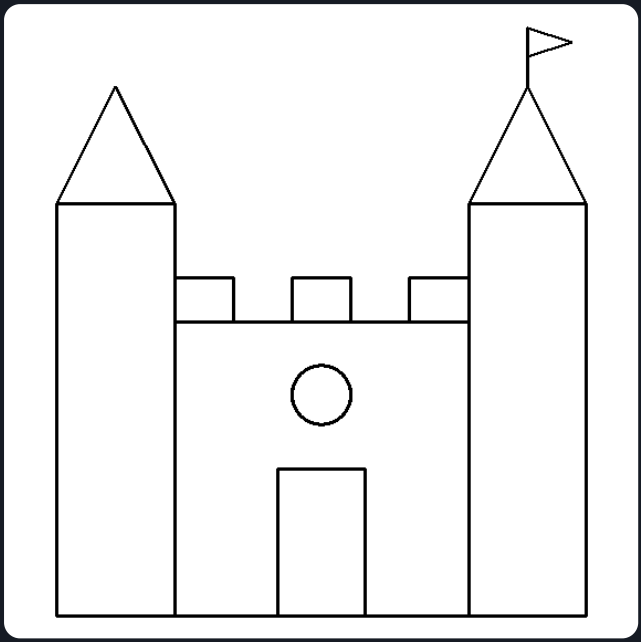
So the object is magnified 3 times, moved by (−1, 4), then rotated 45° around the origin.

$$\begin{vmatrix} \cos 45 & -\sin 45 & 0 \\ \sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

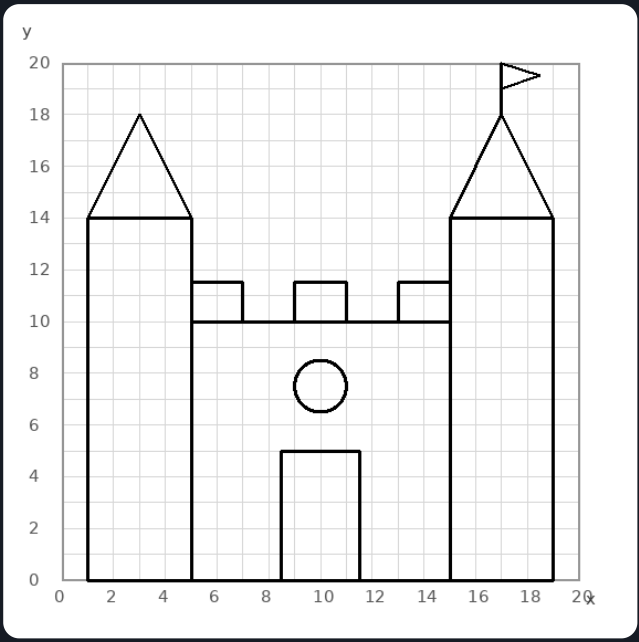
الفكرة اقرا الـ matrices من اليمين للشمال، وكل matrix = جملة بتوصف اللي حصل للشكل.

Q3 Source: main / myInit: Lec 3 · p10 · glBegin / glVertex2f: Lec 3 · p11 · drawCircle (x = xc + r cos θ): Lec 5 · p6–8

Question 3 — program (Castle)



as in the exam



on the 0–20 grid

اكتب الـ includes و drawRect و drawCircle و myInit و main زي صفحة البرنامج الكامل (صفحة 14) — وغيّر بس myDisplay:

myDisplay() START · line 1

```
1  void myDisplay()
2  {
3      glClear(GL_COLOR_BUFFER_BIT);
4      glColor3f(0.0, 0.0, 0.0);
5
6      // left tower
7      drawRect(1, 0, 4, 14, GL_LINE_LOOP);
8
9      // right tower
10     drawRect(15, 0, 4, 14, GL_LINE_LOOP);
11
12     // wall
13     drawRect(5, 0, 10, 10, GL_LINE_LOOP);
14
15     // battlements
16     drawRect(5, 10, 2, 1.5, GL_LINE_LOOP);
17     drawRect(9, 10, 2, 1.5, GL_LINE_LOOP);
18     drawRect(13, 10, 2, 1.5, GL_LINE_LOOP);
19
20     // tower roofs
21     glBegin(GL_LINE_LOOP);
22     glVertex2f(1, 14);
23     glVertex2f(3, 18);
24     glVertex2f(5, 14);
25     glEnd();
```

```
26      glBegin(GL_LINE_LOOP);
27      glVertex2f(15, 14);
28      glVertex2f(17, 18);
29      glVertex2f(19, 14);
30      glEnd();
31
32      // door
33      drawRect(8.5, 0, 3, 5, GL_LINE_LOOP);
34
35      // round window
36      drawCircle(10, 7.5, 1, 1000, 0, GL_LINE_LOOP);
37
38      // flag
39      glBegin(GL_LINES);
40      glVertex2f(17, 18);
41      glVertex2f(17, 20);
42      glEnd();
43      glBegin(GL_LINE_LOOP);
44      glVertex2f(17, 20);
45      glVertex2f(17, 19);
46      glVertex2f(18.5, 19.5);
47      glEnd();
48
49      glFlush();
50  }
```

END · line 50

PART	CODE	إزاي جت الأرقام
left tower	`drawRect(1, 0, 4, 14, …)`	يبدأ من الركن اللي تحت شمال (1, 0) — عرضه 4 وطوله 14 (لحد (5, 14)).
right tower	`drawRect(15, 0, 4, 14, …)`	يبدأ من الركن اللي تحت شمال (15, 0) — عرضه 4 وطوله 14 (لحد (19, 14)).
wall	`drawRect(5, 0, 10, 10, …)`	يبدأ من الركن اللي تحت شمال (5, 0) — عرضه 10 وطوله 10 (لحد (15, 10)).
battlements	`drawRect(5, 10, 2, 1.5, …)`	يبدأ من الركن اللي تحت شمال (5, 10) — عرضه 2 وطوله 1.5 (لحد (7, 11.5)).
battlements	`drawRect(9, 10, 2, 1.5, …)`	يبدأ من الركن اللي تحت شمال (9, 10) — عرضه 2 وطوله 1.5 (لحد (11, 11.5)).
battlements	`drawRect(13, 10, 2, 1.5, …)`	يبدأ من الركن اللي تحت شمال (13, 10) — عرضه 2 وطوله 1.5 (لحد (15, 11.5)).
tower roofs	`glBegin(GL_LINE_LOOP)` · 3 `glVertex2f`	شكل مقفول بالنقط: (1, 14) ← (3, 18) ← (5, 14).
tower roofs	`glBegin(GL_LINE_LOOP)` · 3 `glVertex2f`	شكل مقفول بالنقط: (15, 14) ← (17, 18) ← (19, 14).
door	`drawRect(8.5, 0, 3, 5, …)`	يبدأ من الركن اللي تحت شمال (8.5, 0) — عرضه 3 وطوله 5 (لحد (11.5, 5)).
round window	`drawCircle(10, 7.5, 1, 1000, 0, …)`	دايرة: المركز (10, 7.5) ونص القطر 1.

flag	`glBegin(GL_LINES)` · 2 `glVertex2f`	خطوط بالنقط: (17, 18) ← (17, 20).
flag	`glBegin(GL_LINE_LOOP)` · 3 `glVertex2f`	شكل مقفول بالنقط: (17, 20) ← (19, 17) ← (18.5, 19.5).

الفكرة كل جزء في الصورة = نداء واحد: مستطيل ← drawRect، دايرة ← drawCircle، أي شكل تاني ← glBegin بالنقط.

CG / MODEL ANSWER / MODEL 13

Model Answer — Model 13

M13

Exam paper on page 08 · every answer shows the reference page of its method

Q1-a Source: the page is next to each answer

Question 1 (a) — choose the correct answer

#	ANSWER · WHY	SOURCE
1) A	**Locator** Locator (interactive drawing): coordinate position (x, y).	Lec 1 · p13
2) C	**64** $2^2 \times 2^2 \times 2^2 = 4 \times 4 \times 4 = 64$ colors	Lec 2 · p35
3) A	**(20, 20)** Sx = 500/50 = 10, Sy = 250/50 = 5 · XW = 200/10 = 20 · YW = 100/5 = 20	Graphics_Sec 4 · p2
4) C	**decrement y, calculate x** Region 1 increment x, calculate y · Region 2 decrement y, calculate x.	Lec 5 · p17
5) B	**(1,6) (2,6) (3,5)** P0 = −5 → (1, 6) · P = −2 → (2, 6) · P = 3 ≥ 0 → (3, 5)	Lec 5 · p12

الفكرة المسائل الصغيرة (LUT)، window/viewport، (DDA اتحلّت بنفس قوانين المحاضرة — الأرقام بس جديدة.

Q1-b Source: the page is next to each answer

Question 1 (b) — theory

1 · Define

Input primitives: Different types of data retrieved from input devices. Lec 1 · p13

Homogeneous coordinates: To make a uniform representation to all transformation matrices, add an extra coordinate W (weight = 1) to a point: P(x, y) → P(x, y, 1). Lec 6 · p15

2 · Compare Region 1 and region 2 of the midpoint ellipse Lec 5 · p17, p20–21

Region 1	Region 2
Increment x, calculate y	Decrement y, calculate x
Initial point $(0, r_y)$, $p1_0 = r_y^2 - r_x^2 r_y + \frac{1}{4} r_x^2$	Initial point = last point of region 1, $p2_0 = r_y^2(x_0 + \frac{1}{2})^2 + r_x^2(y_0 - 1)^2 - r_x^2 r_y^2$

Until $2r_y^2x \geq 2r_x^2y$	Repeat until we reach the point $(r_x, 0)$

3 · True / False

1. ✓ **True** A point (x, y) on an ellipse in one quadrant gives the ellipse points in the other three quadrants. Lec 5 · p16
2. ✗ **False** The inverse rotation $R^{-1}(\theta) = R(\theta)$.
   Correct: $R^{-1}(\theta) = R(-\theta) = R^T(\theta)$.
   Lec 6 · p19
3. ✓ **True** In a simple color frame buffer each bit plane drives an individual color gun. Lec 2 · p33

4 · Complete

1. To scale about a point that is not the origin we use translate – _____ – translate. → **scale** Lec 6 · p29
2. In a raster scan system the _____ takes care of the actual display process. → **scan controller** Lec 2 · p30

الفكرة التعريف بجملة المحاضرة، والمقارنة جدول، وأي False لازم تكتب تصحيحه.

Q2-1 Source: Lec 4 · p28–30 · Graphics sec5 · p2

Question 2 (1) — Bresenham line · (5, 5) and (17, 12)

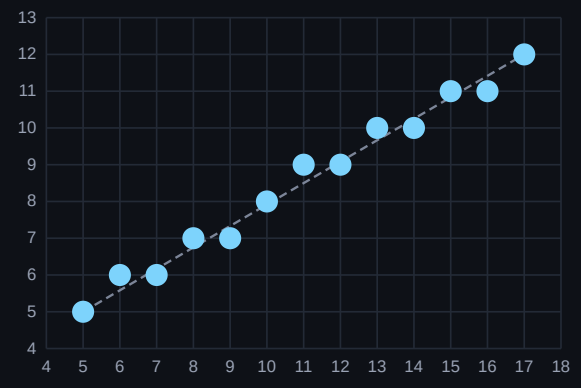
HOW TO SOLVE

1. Compute Δx, Δy and p0 = 2Δy − Δx.
2. Find the two update values: + 2Δy and + 2Δy − 2Δx.
3. Fill the table: the sign of P decides if y increases.
4. Plot the pixels.

$|\Delta x| = |17 - 5| = 12$, $|\Delta y| = |12 - 5| = 7$ · $|\Delta x| > |\Delta y|$ · $p0 = 2\Delta y - \Delta x = 14 - 12 = 2$ · $P_k < 0$: x incremented, y remains, $P_{k+1} = P_k + 14$ · $P_k \geq 0$: x and y incremented, $P_{k+1} = P_k - 10$

P	X	Y	SetPixel(x, y)
-	5	5	(5,5)
2	6	6	(6,6)
−8	7	6	(7,6)
6	8	7	(8,7)
−4	9	7	(9,7)
10	10	8	(10,8)
0	11	9	(11,9)
−10	12	9	(12,9)
4	13	10	(13,10)
−6	14	10	(14,10)

Plotted pixels



8	15	11	(15,11)
−2	16	11	(16,11)
12	17	12	(17,12)

الفكرة الـ P السالب يخلّي y ثابت، والموجب (أو الصفر) يزوّد y بواحد.

Q2-2

Source: Lec 6 · p7, p20–29 · Lec 1 · p22

Question 2 (2) — P′ = T(4, 4) · S(−1, 1) · S(2, 2) · T(−4, −4) · P

Read from right to left — the matrix nearest to P is applied first:

1. **T(−4, −4):** Translate the object so that the fixed point (4, 4) moves to the origin.
2. **S(2, 2):** Scale around the origin: double the size.
3. **S(−1, 1):** Scale with sx = −1: x → −x — a vertical reflection (mirror image across the y-axis).
4. **T(4, 4):** Translate the object so that the fixed point is back to its original position.

Scales multiply: S(−1, 1)·S(2, 2) = S(−2, 2). So the object is doubled and mirrored, keeping the point (4, 4) fixed.

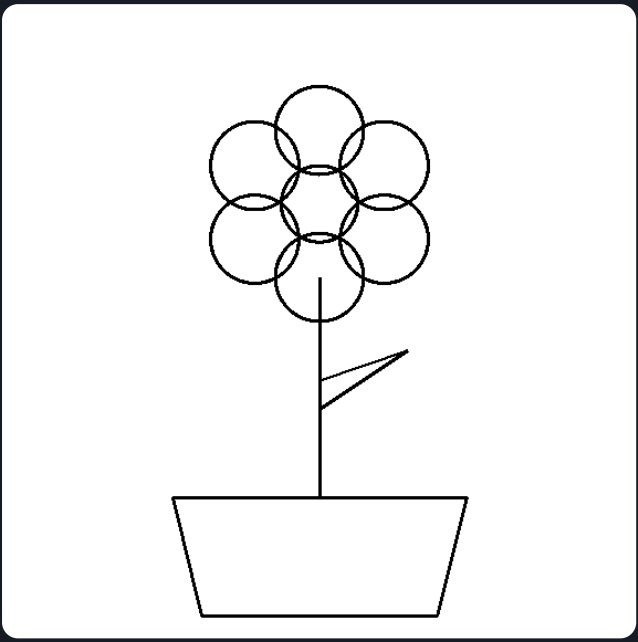
$$\begin{vmatrix} 1 & 0 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} 1 & 0 & -4 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{vmatrix}$$

الفكرة اقرا الـ matrices من اليمين للشمال، وكل matrix = جملة بتوصف اللي حصل للشكل.

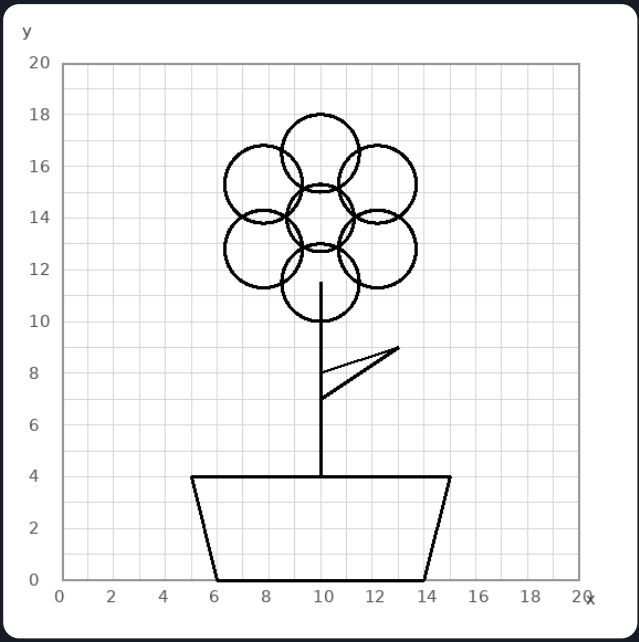
Q3

Source: main / myInit: Lec 3 · p10 · glBegin / glVertex2f: Lec 3 · p11 · drawCircle (x = xc + r cos θ): Lec 5 · p6–8

Question 3 — program (Flower)



as in the exam



on the 0–20 grid

اكتب الـ includes و drawRect و drawCircle و myInit و main زي صفحة البرنامج الكامل (صفحة 14) — وغيّر بس myDisplay:

myDisplay() START · line 1

```
1  void myDisplay()
2  {
3      glClear(GL_COLOR_BUFFER_BIT);
4      glColor3f(0.0, 0.0, 0.0);
5
6      // pot
7      glBegin(GL_LINE_LOOP);
8      glVertex2f(6, 0);
9      glVertex2f(14, 0);
10     glVertex2f(15, 4);
11     glVertex2f(5, 4);
12     glEnd();
13
14     // stem
15     glBegin(GL_LINES);
16     glVertex2f(10, 4);
17     glVertex2f(10, 11.5);
18     glEnd();
19
20     // leaf
21     glBegin(GL_LINE_LOOP);
22     glVertex2f(10, 7);
23     glVertex2f(13, 9);
24     glVertex2f(10, 8);
25     glEnd();
```

```
26
27      // petals
28      drawCircle(10, 16.5, 1.5, 1000, 0, GL_LINE_LOOP);
29      drawCircle(12.2, 15.3, 1.5, 1000, 0, GL_LINE_LOOP);
30      drawCircle(12.2, 12.8, 1.5, 1000, 0, GL_LINE_LOOP);
31      drawCircle(10, 11.5, 1.5, 1000, 0, GL_LINE_LOOP);
32      drawCircle(7.8, 12.8, 1.5, 1000, 0, GL_LINE_LOOP);
33      drawCircle(7.8, 15.3, 1.5, 1000, 0, GL_LINE_LOOP);
34
35      // center
36      drawCircle(10, 14, 1.3, 1000, 0, GL_LINE_LOOP);
37
38      glFlush();
39  }
```

END · line 39

PART	CODE	إزاي جت الأرقام
pot	glBegin(GL_LINE_LOOP) · 4 glVertex2f	شكل مقفول بالنقط: (6, 0) ← (14, 0) ← (15, 4) ← (5, 4).
stem	glBegin(GL_LINES) · 2 glVertex2f	خطوط بالنقط: (10, 4) ← (10, 11.5).
leaf	glBegin(GL_LINE_LOOP) · 3 glVertex2f	شكل مقفول بالنقط: (10, 7) ← (13, 9) ← (10, 8).
petals	drawCircle(10, 16.5, 1.5, 1000, 0, …)	دايرة: المركز (10, 16.5) ونص القطر 1.5.
petals	drawCircle(12.2, 15.3, 1.5, 1000, 0, …)	دايرة: المركز (12.2, 15.3) ونص القطر 1.5.
petals	drawCircle(12.2, 12.8, 1.5, 1000, 0, …)	دايرة: المركز (12.2, 12.8) ونص القطر 1.5.
petals	drawCircle(10, 11.5, 1.5, 1000, 0, …)	دايرة: المركز (10, 11.5) ونص القطر 1.5.
petals	drawCircle(7.8, 12.8, 1.5, 1000, 0, …)	دايرة: المركز (7.8, 12.8) ونص القطر 1.5.
petals	drawCircle(7.8, 15.3, 1.5, 1000, 0, …)	دايرة: المركز (7.8, 15.3) ونص القطر 1.5.
center	drawCircle(10, 14, 1.3, 1000, 0, …)	دايرة: المركز (10, 14) ونص القطر 1.3.

الفكرة كل جزء في الصورة = نداء واحد: مستطيل ← drawRect، دايرة ← drawCircle، أي شكل تاني ← glBegin بالنقط.

CG / MODEL ANSWER / MODEL 14

Model Answer — Model 14

M14

Exam paper on page 10 · every answer shows the reference page of its method

Q1-a

Source: the page is next to each answer

Question 1 (a) — choose the correct answer

#	ANSWER · WHY	SOURCE
1) C	**shifted slices** Shear: deform shape like shifted slices.	Lec 6 · p11
2) B	**64** $2^6 = 64$ gray levels	Lec 2 · p23
3) D	**(100, 50)** Sx = 200/100 = 2, Sy = 100/50 = 2 · XV = 0 + 2(0 + 50) = 100 · YV = 0 + 2(0 + 25) = 50	Lec 3 · p5–6
4) C	**$S(1/s_x, 1/s_y)$** $S^{-1}(s_x, s_y) = S(1/s_x, 1/s_y)$	Lec 6 · p24
5) B	**(1,0) (2,1) (3,1)** Δx = 8, Δy = 3 · p0 = −2 → (1, 0) · p = 4 → (2, 1) · p = −6 → (3, 1)	Lec 4 · p28

الفكرة المسائل الصغيرة (LUT)، window/viewport، (DDA اتحلّت بنفس قوانين المحاضرة — الأرقام بس جديدة.

Q1-b

Source: the page is next to each answer

Question 1 (b) — theory

1 · Define

Display device: An output device used to represent the information in visual form. Lec 2 · p26

Color depth: The number of bits used for the color of a pixel; true color = 24 bit (8 for each color). Lec 2 · p24

2 · Compare Vertex shader and fragment shader Lec 1 · p24

Vertex shader	Fragment shader
Defines object vertices	Adds values for each pixel
Moves the object (translation, scaling, rotation)	Colors the object shape

First stage of the pipeline	Last stage (after rasterization)

3 · True / False

1. ✓ **True** Scaling preserves the angles between lines only with uniform scaling $s_x = s_y$. Lec 6 · p6
2. ✓ **True** In old raster scan systems the electron beam is swept across the screen one row at a time. Lec 2 · p29
3. ✗ **False** The midpoint circle algorithm computes one quadrant and gets the rest by symmetry.

   Correct: It computes one octant; the other seven come from the eight-way symmetry.

   Lec 5 · p10, p13

4 · Complete

1. Geometric transformation functions: translation, scaling, rotation, _____ and _____. → **reflection · shearing**

   Lec 6 · p2
2. Diagonal reflection: x′ = _____, y′ = _____. → **−x · −y** Lec 6 · p7

الفكرة التعريف بجملة المحاضرة، والمقارنة جدول، وأي False لازم تكتب تصحيحه.

Q2-1 Source: Lec 5 · p16–23 · Graphics_Sec 6 · p2

Question 2 (1) — Midpoint ellipse · $r_x = 6$ and $r_y = 2$

HOW TO SOLVE

1. Compute r_x^2, r_y^2 and $p1_0$.
2. Region 1: fill the table (increment x) until $2r_y^2x \geq 2r_x^2y$.
3. Compute $p2_0$ from the last point of region 1.
4. Region 2: fill the table (decrement y) until y = 0.
5. Plot the points.

$r_x^2 = 36$, $r_y^2 = 4$ · initial point (0, 2)

$p1_0 = r_y^2 - r_x^2 r_y + \frac{1}{4}r_x^2 = 4 - 36\times2 + \frac{1}{4}\times36 = -59$

Region 1

Pk_region1	x	y	$2r_y^2x$	$2r_x^2y$
	0	2	0	144
−59	1	2	8	144
−47	2	2	16	144
−27	3	2	24	144
1	4	1	32	72
−35	5	1	40	72
9	6	0	48	0

Region 1 stops when $2r_y^2x \geq 2r_x^2y$ → at (6, 0): $48 \geq 0$

Initial point of region 2 = last point of region 1 (6, 0) · $p2_0 = r_y^2(x_0 + \frac{1}{2})^2 + r_x^2(y_0 - 1)^2 - r_x^2r_y^2 = 4\times(6.5)^2 + 36\times(-1)^2 - 36\times4 = 61$

Region 2

Pk_region2	x	y	$2r_y^2x$	$2r_x^2y$
	6	0	48	0

Region 1 (blue) · Region 2 (pink)



الفكرة Region 1 بنزوّد x لحد ما x_y^2r يبقى ≤ y_x^2r، وبعدين Region 2 بنقلّل y لحد الصفر.

Q2-2

Source: Lec 6 · p7, p20–29 · Lec 1 · p22

Question 2 (2) — P′ = T(2, 3) · R(60) · T(−2, −3) · T(6, 0) · P

Read from right to left — the matrix nearest to P is applied first:

1. **T(6, 0):** Translate the object 6 units to the right.
2. **T(−2, −3):** Translate so that the point (2, 3) moves to the origin.
3. **R(60):** Rotate 60 degree counter-clockwise around the origin.
4. **T(2, 3):** Translate the point (2, 3) back to its original position.

So the object is moved 6 units right, then rotated 60° around the point (2, 3). (Translations add: T(−2, −3)·T(6, 0) = T(4, −3).)

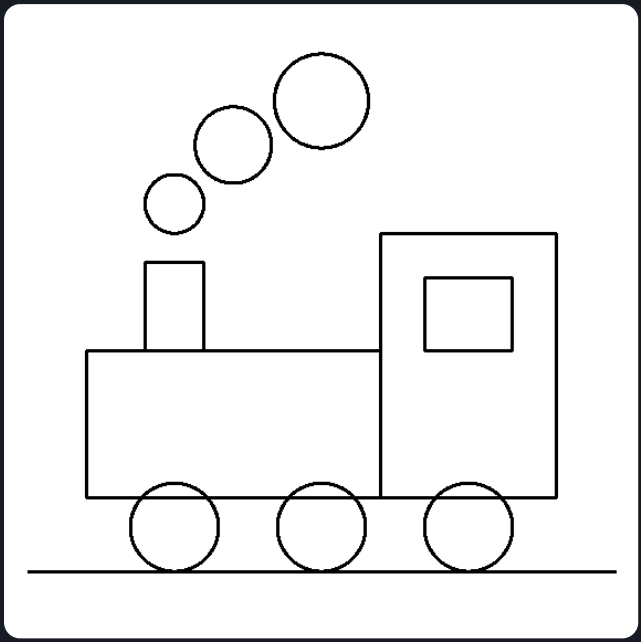
$$\begin{vmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} \cos 60 & -\sin 60 & 0 \\ \sin 60 & \cos 60 & 0 \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} 1 & 0 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

الفكرة اقرا الـ matrices من اليمين للشمال، وكل matrix = جملة بتوصف اللي حصل للشكل.

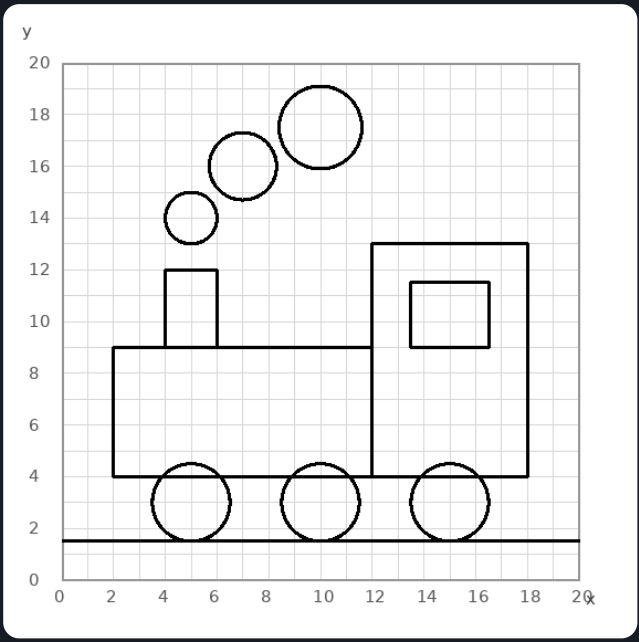
Q3

Source: main / myInit: Lec 3 · p10 · glBegin / glVertex2f: Lec 3 · p11 · drawCircle (x = xc + r cos θ): Lec 5 · p6–8

Question 3 — program (Train)



as in the exam



on the 0–20 grid

اكتب الـ includes و drawRect و drawCircle و myInit و main زي صفحة البرنامج الكامل (صفحة 14) — وغيّر بس myDisplay:

myDisplay() START · line 1

```
1  void myDisplay()
2  {
3      glClear(GL_COLOR_BUFFER_BIT);
4      glColor3f(0.0, 0.0, 0.0);
5
6      // engine body
7      drawRect(2, 4, 10, 5, GL_LINE_LOOP);
8
9      // cab
10     drawRect(12, 4, 6, 9, GL_LINE_LOOP);
11
12     // cab window
13     drawRect(13.5, 9, 3, 2.5, GL_LINE_LOOP);
14
15     // chimney
16     drawRect(4, 9, 2, 3, GL_LINE_LOOP);
17
18     // smoke
19     drawCircle(5, 14, 1, 1000, 0, GL_LINE_LOOP);
20     drawCircle(7, 16, 1.3, 1000, 0, GL_LINE_LOOP);
21     drawCircle(10, 17.5, 1.6, 1000, 0, GL_LINE_LOOP);
22
23     // wheels
24     drawCircle(5, 3, 1.5, 1000, 0, GL_LINE_LOOP);
25     drawCircle(10, 3, 1.5, 1000, 0, GL_LINE_LOOP);
```

```
26      drawCircle(15, 3, 1.5, 1000, 0, GL_LINE_LOOP);
27
28      // rail
29      glBegin(GL_LINES);
30      glVertex2f(0, 1.5);
31      glVertex2f(20, 1.5);
32      glEnd();
33
34      glFlush();
35  }
```

END · line 35

PART	CODE	إزاي جت الأرقام
engine body	`drawRect(2, 4, 10, 5, …)`	يبدأ من الركن اللي تحت شمال (2, 4) — عرضه 10 وطوله 5 (لحد (12, 9)).
cab	`drawRect(12, 4, 6, 9, …)`	يبدأ من الركن اللي تحت شمال (12, 4) — عرضه 6 وطوله 9 (لحد (18, 13)).
cab window	`drawRect(13.5, 9, 3, 2.5, …)`	يبدأ من الركن اللي تحت شمال (13.5, 9) — عرضه 3 وطوله 2.5 (لحد (16.5, 11.5)).
chimney	`drawRect(4, 9, 2, 3, …)`	يبدأ من الركن اللي تحت شمال (4, 9) — عرضه 2 وطوله 3 (لحد (6, 12)).
smoke	`drawCircle(5, 14, 1, 1000, 0, …)`	دايرة: المركز (5, 14) ونص القطر 1.
smoke	`drawCircle(7, 16, 1.3, 1000, 0, …)`	دايرة: المركز (7, 16) ونص القطر 1.3.
smoke	`drawCircle(10, 17.5, 1.6, 1000, 0, …)`	دايرة: المركز (10, 17.5) ونص القطر 1.6.
wheels	`drawCircle(5, 3, 1.5, 1000, 0, …)`	دايرة: المركز (5, 3) ونص القطر 1.5.
wheels	`drawCircle(10, 3, 1.5, 1000, 0, …)`	دايرة: المركز (10, 3) ونص القطر 1.5.
wheels	`drawCircle(15, 3, 1.5, 1000, 0, …)`	دايرة: المركز (15, 3) ونص القطر 1.5.
rail	`glBegin(GL_LINES) · 2 glVertex2f`	خطوط بالنقط: (0, 1.5) ← (20, 1.5).

الفكرة كل جزء في الصورة = نداء واحد: مستطيل ← drawRect، دايرة ← drawCircle، أي شكل تاني ← glBegin بالنقط.

CG / MODEL ANSWER / MODEL 15

Model Answer — Model 15

M15

Exam paper on page 12 · every answer shows the reference page of its method

Q1-a

Source: the page is next to each answer

Question 1 (a) — choose the correct answer

#	ANSWER · WHY	SOURCE
1) C	**device independence** To ensure device independence, graphics systems convert object descriptions to normalized coordinates.	Lec 3 · p2
2) A	**512 entries, 1024 bytes** 2^9 = 512 entries · (512 × 16)/8 = 1024 bytes	Lec 2 · p39
3) C	**(50, 75)** Sx = Sy = 100/20 = 5 · XV = 5(20 − 10) = 50 · YV = 5(25 − 10) = 75	Lec 3 · p5–6
4) A	**R(90)** R(θ) · R(φ) = R(θ + φ) → R(90)	Lec 6 · p26
5) C	**−59** $p1_0 = r_y^2 - r_x^2 r_y + \frac{1}{4} r_x^2 = 4 - 72 + 9 = -59$	Lec 5 · p20

الفكرة المسائل الصغيرة (LUT)، window/viewport، (DDA اتحلّت بنفس قوانين المحاضرة — الأرقام بس جديدة.

Q1-b

Source: the page is next to each answer

Question 1 (b) — theory

1 · Define

Frame rate and refresh rate: Frame rate (FPS): complete images the GPU sends each second · Refresh rate (Hz): times per second the monitor can physically update the image. Lec 2 · p28

Composite transformation: Combine transformations of different type (translate, rotate, translate · translate, scale, translate · translate, reflect, translate), implemented by multiplication of the corresponding homogeneous matrices. Lec 6 · p29

2 · Compare Direct (Pythagoras) circle method and midpoint circle Lec 5 · p5, p11–13

$y = yc \pm \sqrt{(r^2 - (x - xc)^2)}$	Midpoint circle

Method needs lots of computation	Incremental decision parameter (P0 = 1 − r)
Gives non-uniform pixel spacing	Only one octant's calculation needed (eight-way symmetry)

3 · True / False

1. ✗ **False** A valuator gives a coordinate position (x, y).
   Correct: That is the locator; the valuator gives scalar values (typically between 0 and 1).
   Lec 1 · p13
2. ✓ **True** $T(t_{x1}, t_{y1}) \cdot T(t_{x2}, t_{y2}) = T(t_{x1} + t_{x2}, t_{y1} + t_{y2})$. Lec 6 · p20–22
3. ✓ **True** For a line with slope greater than 1 in Bresenham, we interchange the roles of x and y.
   Graphics sec5 · p1 · Lec 4 · p31

4 · Complete

1. glutInitWindowSize(400, 300) creates a display window with a width of _____ and a height of _____. → **400 pixels · 300 pixels** Lec 1 · p21
2. Supersampling gives more weight to subpixels near the _____ of a pixel area. → **center** Lec 5 · p25

الفكرة التعريف بجملة المحاضرة، والمقارنة جدول، وأي False لازم تكتب تصحيحه.

Q2-1

Source: Lec 4 · p19–23 · Graphics sec5 · p1

Question 2 (1) — DDA line · (3, 1) to (7, 10)

HOW TO SOLVE

1. Compute Δx, Δy, steps and m (or 1/m).
2. Decide the increments (+ or −) from the direction of the line.
3. Add the increments step by step and round each value.
4. Plot the pixels.

Δx = 4, Δy = 9 → |Δy| > |Δx| (|m| > 1) · 1/m = 0.44 · Steps = 9 · $y_{k+1} = y_k + 1$ · $x_{k+1} = x_k + 1/m$

K	x	y	(x, y)
0	3.44	2	(3,2)
1	3.89	3	(4,3)
2	4.33	4	(4,4)
3	4.78	5	(5,5)
4	5.22	6	(5,6)
5	5.67	7	(6,7)
6	6.11	8	(6,8)
7	6.56	9	(7,9)

Plotted pixels

8	7	10	(7,10)



الفكرة الـ coordinate اللي بيتغير أكتر بيمشي بخطوة 1، والتاني بيزيد بالكسر ويتقرّب.

Q2-2

Source: Lec 6 · p7, p20–29 · Lec 1 · p22

Question 2 (2) — P′ = S(1, −1) · T(0, 5) · R(180) · P

Read from right to left — the matrix nearest to P is applied first:

1. **R(180):** Rotate 180 degree around the origin: (x, y) → (−x, −y).
2. **T(0, 5):** Translate the object 5 units up.
3. **S(1, −1):** Scale with sy = −1: y → −y — a horizontal reflection (mirror image across the x-axis).

So the object is turned 180°, moved 5 units up, then mirrored upside-down across the x-axis.

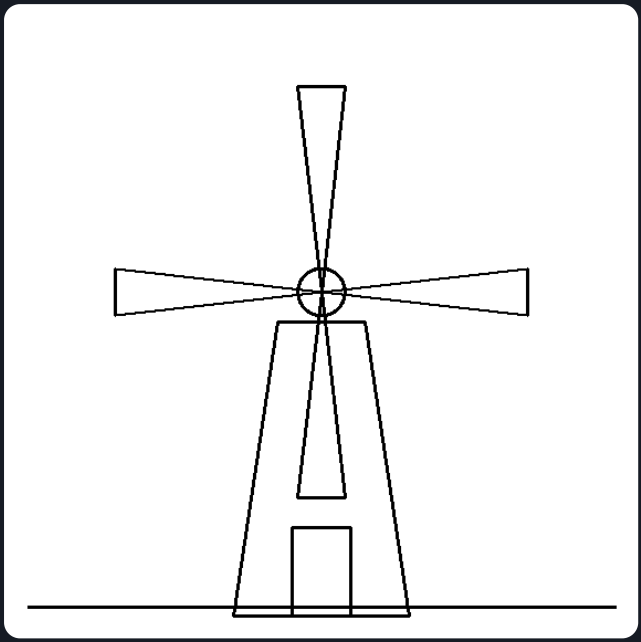
$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} \cos 180 & -\sin 180 & 0 \\ \sin 180 & \cos 180 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

الفكرة اقرا الـ matrices من اليمين للشمال، وكل matrix = جملة بتوصف اللي حصل للشكل.

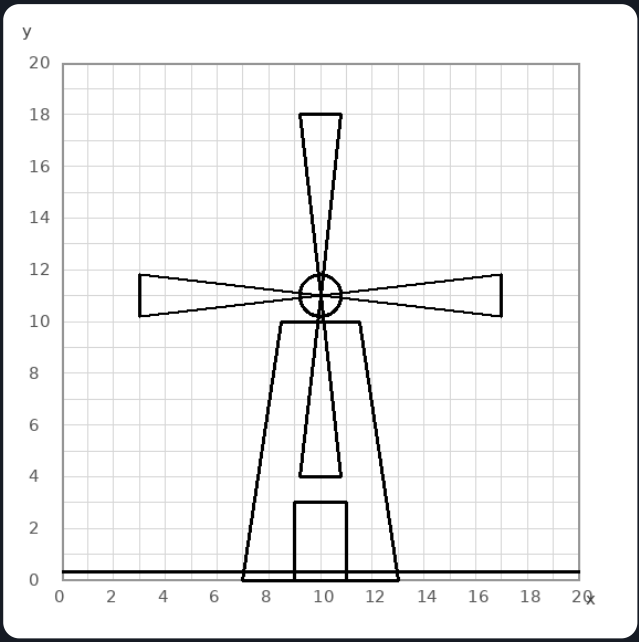
Q3

Source: main / myInit: Lec 3 · p10 · glBegin / glVertex2f: Lec 3 · p11 · drawCircle (x = xc + r cos θ): Lec 5 · p6–8

Question 3 — program (Windmill)



as in the exam



on the 0–20 grid

اكتب الـ includes و drawRect و drawCircle و myInit و main زي صفحة البرنامج الكامل (صفحة 14) — وغيّر بس myDisplay:

myDisplay() START · line 1

```
1  void myDisplay()
2  {
3      glClear(GL_COLOR_BUFFER_BIT);
4      glColor3f(0.0, 0.0, 0.0);
5
6      // ground
7      glBegin(GL_LINES);
8      glVertex2f(0, 0.3);
9      glVertex2f(20, 0.3);
10     glEnd();
11
12     // tower
13     glBegin(GL_LINE_LOOP);
14     glVertex2f(7, 0);
15     glVertex2f(13, 0);
16     glVertex2f(11.5, 10);
17     glVertex2f(8.5, 10);
18     glEnd();
19
20     // door
21     drawRect(9, 0, 2, 3, GL_LINE_LOOP);
22
23     // blades
24     glBegin(GL_LINE_LOOP);
25     glVertex2f(10, 11);
```

```
    glVertex2f(9.2, 18);
    glVertex2f(10.8, 18);
    glEnd();
    glBegin(GL_LINE_LOOP);
    glVertex2f(10, 11);
    glVertex2f(17, 11.8);
    glVertex2f(17, 10.2);
    glEnd();
    glBegin(GL_LINE_LOOP);
    glVertex2f(10, 11);
    glVertex2f(3, 11.8);
    glVertex2f(3, 10.2);
    glEnd();
    glBegin(GL_LINE_LOOP);
    glVertex2f(10, 11);
    glVertex2f(9.2, 4);
    glVertex2f(10.8, 4);
    glEnd();

    // hub
    drawCircle(10, 11, 0.8, 1000, 0, GL_LINE_LOOP);

    glFlush();
}
```

END · line 49

PART	CODE	إزاي جت الأرقام
ground	`glBegin(GL_LINES) · 2 glVertex2f`	خطوط بالنقط: (0, 0.3) ← (20, 0.3).
tower	`glBegin(GL_LINE_LOOP) · 4 glVertex2f`	شكل مقفول بالنقط: (7, 0) ← (13, 0) ← (11.5, 10) ← (8.5, 10).
door	`drawRect(9, 0, 2, 3, …)`	يبدأ من الركن اللي تحت شمال (9, 0) — عرضه 2 وطوله 3 (لحد (11, 3)).
blades	`glBegin(GL_LINE_LOOP) · 3 glVertex2f`	شكل مقفول بالنقط: (10, 11) ← (9.2, 18) ← (10.8, 18).
blades	`glBegin(GL_LINE_LOOP) · 3 glVertex2f`	شكل مقفول بالنقط: (10, 11) ← (11.8, 17) ← (10.2, 17).
blades	`glBegin(GL_LINE_LOOP) · 3 glVertex2f`	شكل مقفول بالنقط: (10, 11) ← (11.8, 3) ← (10.2, 3).
blades	`glBegin(GL_LINE_LOOP) · 3 glVertex2f`	شكل مقفول بالنقط: (10, 11) ← (9.2, 4) ← (10.8, 4).
hub	`drawCircle(10, 11, 0.8, 1000, 0, …)`	دايرة: المركز (10, 11) ونص القطر 0.8.

الفكرة كل جزء في الصورة = نداء واحد: مستطيل ← drawRect، دايرة ← drawCircle، أي شكل تاني ← glBegin بالنقط.

CG / PART 4 / QUICK ANSWER KEY

Quick Answer Key

Question 1 of all five models, for fast checking

K

MODEL	MCQ 1–5	T / F	COMPLETE	PAGE
Model 11	C A C B B	✗ ✓ ✗	selection of a part of the screen | 256	17
Model 12	B D B D C	✗ ✓ ✗	$x_k + 1$ · $y_k + m$ | enlarged	23
Model 13	A C A C B	✓ ✗ ✓	scale | scan controller	29
Model 14	C B D C B	✓ ✓ ✗	reflection · shearing | −x · −y	34
Model 15	C A C A C	✗ ✓ ✓	400 pixels · 300 pixels | center	39

Score yourself

- Question 1: 2.5 points for the MCQ (0.5 each) + 2.5 points for part b.
- Question 2: 5 points for the algorithm table + plot, 5 points for the step-by-step description.
- Question 3: 5 points — the program template + a myDisplay that draws every part of the figure.

الدرجات دي تقسيمة مني للتدريب بس — مش من الدكتورة.